



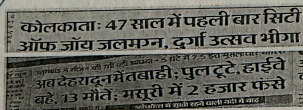
Malaviya National Institute of Technology Jaipur
I Semester B.Tech. (Section A & D) Mid-term Examination 2025-26
22CET102: Environmental Science (Course Instructor: Dr. Nivedita Kaul)

Max. Marks: 110

Time Allowed: 1½ Hours

Note: Answer all questions in points. Draw a line after each answer and mark 'The End' at the last.

1.



पंजाब में 37 साल के बाद सबसे भीषण बाढ़

Explain the reasons for these extreme weather events. How such events can be avoided/reduced? 10+10=20

2. a) 'Meteorology plays an important role in diluting/dispersing air pollutants'. Explain giving examples of IOC fire or Bhopal Gas Tragedy. 10
- b) What parameters influence the indoor environment quality? 10
3. a) 'Noise is a modern plague'. List 10 reasons to justify the statement. As per the Indian standards, what is the acceptable noise level for MNIT campus during night? 10
- b) When does sound become noise? Sound pressure level scale is similar to the pH or Richter Scale. Explain. Calculate SPL of a sound with pressure 0.02 Pascal. $SPL = 20 \log_{10} \left(\frac{P}{P_{ref}} \right)$ 10
4. a) Why is emission of NO_x undesirable? If the AQI of Delhi reaches 450, what are its implications and how will the government respond? 10
- b) What is the 'Earth Overshooting Day'? List 5 reasons, you think it came on July 25 this year. 10
5. a) Name any 10 SDGs. 5
- b) On a map of world, name and locate any 5 places where COPs have been held recently. On a map of India, name and locate 5 places witnessing extreme weather events in last few months. (Exclude names given in this question paper) 5
6. Answer in short 2½ X 4 = 10
- a) Role of CPCB b) Role of MoEF&CC c) Role of IPCC d) Relevance of Climate clock
7. Name any two Indian persons and two Indian organizations (excluding those given in this question paper) that work in the field of environment. Mention any two newspaper articles and two videos, shared during the class, that you found informative. 10

$$20 \log_{10} \left(\frac{0.02}{20000} \right)$$

$$20 \log_{10} (10)$$
$$20 \log_{10} \left(\frac{1}{n} \sum_{i=1}^n (10) \right)$$

$$P_i = 20$$
$$L_p$$
$$P_i$$

Malaviya National Institute of Technology Jaipur
Mid-term examination (26th September 2024)

B.Tech first semester (For section C & E) Environmental Science(22CET102)

Time allowed: 1.5 hours

Maximum marks:100

Course Outcomes

- CO-1 To be able to analyse and evaluate aspects of local and global environmental issues and apply understanding to create informed opinions about how to interact with the environment on both a personal and a social level.*
- CO-2 To consider environmental issues as among the highest educational priorities and as a key determinant to sustainable development.*
- CO-3 To sensitize about existing environmental conditions, policies, programs and practices for implementation at different stages of life in an environmentally sound manner.*
- CO-4 To create awareness about various techniques/practices to conserve and protect the environment.*

NOTE: Please return the question paper, uncrumpled. Answer all questions in the same order as asked, neatly in points.

1. Typhoon Yagi has caused severe destruction of life and property in China, Vietnam, Hong Kong, etc. recently. Off late, the intensity and frequency of extreme weather events have increased. explain in detail
 - a) The reasons for this
 - b) Efforts being made on a global scale to avoid such events(20 marks)
2. How is AQI calculated? What is its significance in India? (10 marks)
3. Why is tropospheric ozone undesirable?(10 marks)
4. How is stratospheric ozone formed? Name few scientists and organisations associated with its discovery, destruction and measurements. (10 marks)
5. In a country in South America, an ozone column of 150DU was observed in last week of September, by a NASA satellite. What does this imply/mean for its environment? (10 marks)
6. In a tabular form give a list of commercially available vehicular fuels in India and details of the pollutants they emit (10 marks)
7. Why is the following undesirable?
 - a) Stub burning in Punjab
 - b) Promoting use of HFCs
 - c) Emissions of NO_x
 - d) Emissions of fine PM(20 marks)
8. Write full form of
 - a) IPCC
 - b) CPCB
 - c) MOEF&CC
 - d) NAAQS
 - 1) On a world map locate and name places affected by recent forest fires (give year) (3 marks)
 - 2) On a map of India locate places that experienced massive landslides, floods and cyclones in the past 1 year. (3 marks)

*Please draw a line after each answer
After finishing your paper, write 'The End'*